

Lecture 6 Scribing Notes — Clustering (Page 38 to 69)

Distance Measures, Hierarchical Clustering and Mahalanobis Distance

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1. Minkowski Distance

Minkowski distance is a generalized way to measure the distance between two data points. It puts several common distance metrics into one formula.

For two data points p and q ,

$$d(p, q) = (\sum_{k=1}^m |p_k - q_k|^r)^{\frac{1}{r}}$$

where m = no. of dimensions, and r decides which type of distance is used.

Important special cases:

- $r=1$: Manhattan distance, also called city-block, taxicab, or L_1 distance
- $r=2$: Euclidean distance
- $r \rightarrow \infty$: L_∞ / supremum distance $\rightarrow$ determined by the largest difference in any dimension

For binary vectors, **Hamming distance** is closely related to Manhattan distance $\rightarrow$ counts how many positions are different.

Different r values $\rightarrow$ different ways of measuring distance $\rightarrow$ possibly different distance matrices + clustering results.

2. Distance and Similarity in Clustering

Clustering aims to put similar objects into the same group.

Ideally:

small intra-cluster difference

and

large inter-cluster difference.

Need a way to define how similar / different two objects are.

Common measures:

- cosine similarity
- correlation
- Euclidean distance
- Manhattan distance
- **Mahalanobis distance**

Same dataset + different definition of “similar” $\rightarrow$ **different clustering result.**

3. Hierarchical Clustering

Hierarchical clustering builds clusters step by step and organizes them into a tree structure.

Result usually shown using a **dendrogram** $\rightarrow$ records how clusters are merged. Useful in e.g. gene clustering, phylogeny.

Basic procedure:

1. Compute similarity / distance matrix
2. Treat each data point as one cluster
3. Merge the two closest clusters
4. Update the matrix
5. Repeat until only one cluster remains

find closest pair $\rightarrow$ merge $\rightarrow$ update matrix $\rightarrow$ repeat

After each merge, two individual points become one larger cluster

$\rightarrow$ distance / similarity to remaining clusters must be updated.

4. How to Update the Distance Matrix

After two clusters are merged, their distance to other clusters has to be redefined.

Possible choices:

- **Minimum distance** — closest pair of points
- **Maximum distance** — farthest pair
- **Group average** — avg. pairwise distance
- **Centroid distance** — compare cluster centers

Different update rules → different hierarchical structures.

Distance-based interpretation:

smaller distance $\Rightarrow$ more similar

Correlation-based interpretation:

larger correlation $\Rightarrow$ more similar

$\therefore$ in the gene example:

minimum distance $\leftrightarrow$ largest correlation

Same idea, different measurement.

5. Gene Expression Example

Running example: 5 genes measured under 4 conditions:

- wild type
- mutant 1
- mutant 2
- mutant 3

Q: What should be used to compare the genes?

Absolute expression values may not be the best measure. Two genes can have very different expression levels but still change in a similar way across conditions.

For gene-expression data:

pattern of change > absolute magnitude

e.g.

low $\rightarrow$ high $\rightarrow$ high $\rightarrow$ low

Two genes following the same pattern may still be similar even if one has much higher expression.

$\rightarrow$ use **linear correlation** to compare expression patterns.

6. Correlation as Similarity

Correlation measures how two variables change together.

$$-1 \leq r \leq 1$$

- $r \approx 1$: very similar trend
- $r \approx -1$: opposite trend
- $r \approx 0$: weak linear relationship

Correlation matrix gives:

$$\text{corr}(\text{At1g30720}, \text{At2g34930}) = 0.9909$$

$\rightarrow$ strongest positive correlation among the relevant pairs.

$\therefore$ merge these two genes first.

After merge:

- update correlation matrix
- pick largest remaining correlation
- merge next closest cluster

At4g35770 joins the cluster, followed later by At2g05540.

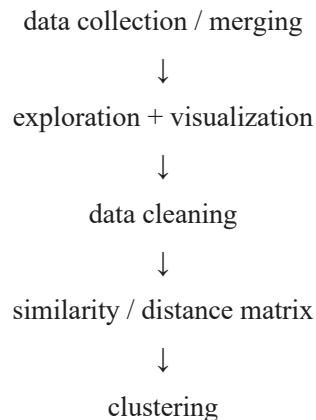
No need to memorize every number in the matrix. Main process:

calculate similarity → merge → update → repeat

Whole merge history → **dendrogram**.

7. Gene Expression Clustering Workflow

The gene-expression example also fits into a larger data-analysis workflow:



Dimension reduction can be added later.

∴ clustering result depends not only on the algorithm, but also on preprocessing + similarity measure.

8. Mahalanobis Distance

Mahalanobis distance is a more advanced distance measure. Instead of only looking at raw numerical differences, it also considers the **data distribution**.

$$d_M(x, y) = \sqrt{(x - y)^T \Sigma^{-1} (x - y)}$$

where Σ = **covariance matrix**.

Compared w/ Euclidean distance, it also considers:

- how much each variable normally varies
- how different variables are related

Same Euclidean distance ≠ same **Mahalanobis distance**.

9. Quiz Example — Why Raw Difference Is Not Enough

Suppose there are two quizzes.

Quiz 1:

std=10

Student A – Student B = 1 mark.

Quiz 2:

std=1

Student C – Student D = 1 mark.

Raw difference:

1=1

but meaning is different.

Quiz 1: 1 mark is small compared w/ normal variation.

Quiz 2: 1 mark is much more noticeable.

Better comparison:

How many standard deviations apart?

→ intuition behind **Mahalanobis distance**.

10. Why Covariance Matters

Mahalanobis distance also considers relationships between variables.

e.g. height and weight are correlated. A change following the usual data pattern may be less unusual than a change in an uncommon direction.

→ distance depends on both **variance + covariance**, not only raw coordinate difference.

11. Mahalanobis Distance and Normalization

This connects back to normalization.

If one variable is measured in meters and another in kilograms, their scales are very different.

Using raw Euclidean distance:

large-scale variable may dominate

One solution → normalization, e.g. min-max normalization. The lecture uses height and weight as an example.

Simple comparison:

Raw Euclidean distance

→ compare numerical differences directly

Normalized Euclidean distance

→ rescale variables first, then compare

Mahalanobis distance

→ **variance + covariance** + overall data distribution

12. Summary

Main flow:

data → similarity / distance → matrix → merge closest clusters → update → repeat → dendrogram

Minkowski distance → different ways to define distance.

Hierarchical clustering → repeatedly merge closest clusters.

Gene-expression clustering → correlation useful when pattern matters more than absolute value.

Mahalanobis distance → considers **variance + covariance**, not just raw difference.